be consumed by livestock. By combining two types of devices, which typically require a large land area each, the system becomes simple and compact. General arrangement of dual-purpose photovoltaic (PV) cells is shown in Fig. 1.

Simultaneous production of fresh water and electricity has been achieved by an integrated solar PV panel - membrane distillation (PV-MD) device. In a typical solar cell, eighty to ninety per cent of absorbed solar energy is undesirably converted to heat and then it is passively and wastefully dumped into ambient air.

PV cells are protected by glass and held together in a metal frame. These are comparable to materials that make up windows and frame of a car. When a car is parked in a parking lot on a hot summer day, it gets extremely hot, but there is little danger of it burning. PV cells can get as hot as 65°C in the summer if there is no proper ventilation. PVs are generally tested at 25°C and rated at that temperature. If a panel is rated to have a temperature coefficient of 0.50 per cent per degree Celsius, then PV power will decrease by half a per cent every degree the temperature rises above 25°C.

In the laboratory it has been found that, when a PV panel reaches maximum allowable temperature of 45°C, air cooling by a fan decreases the temperature by 4.7°C and, hence, efficiency is increased by 2.6 per cent. But if the same panel is cooled by heat pipes using water as in MD, temperature reduces by 8°C and PV efficiency increases by more than three per cent.

By mounting a water distillation system on the back of a solar cell, engineers have constructed a device that doubles as an energy generator and water purifier. While the solar cells harvest sunlight for electricity, heat from the solar panels drives evaporation in the water distiller below. Vapour wafts through a porous polystyrene membrane that filters out salt and other contaminants, allowing clean water to condense on the other side. This does not affect electricity production and at the same time gives fresh water, deemed safe without dissolved salts, suspended materials and free from any bacteria, as bonus.

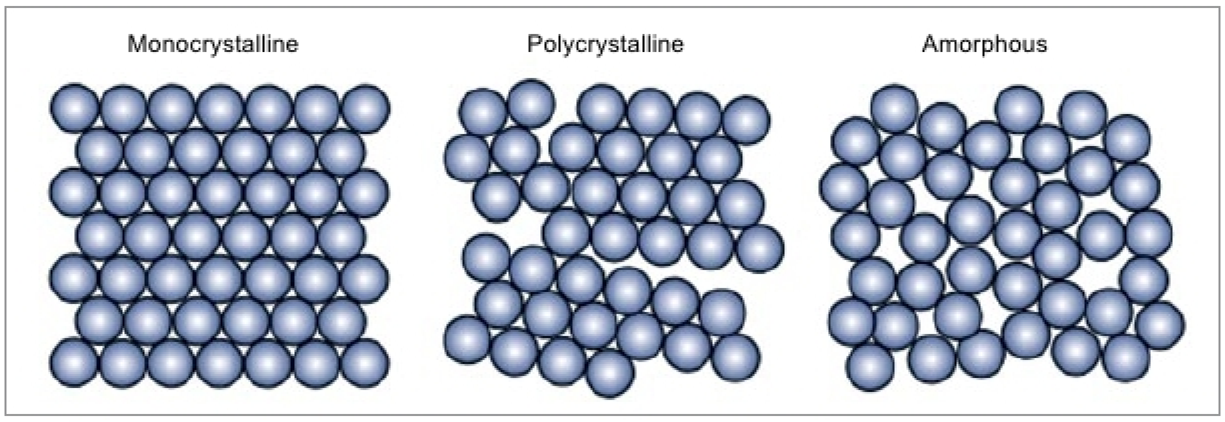

Fig. 2: Various forms of silicon crystals

Selection of PV module

Crystalline solar cells have dominated the PV market since the 1950s. Silicon is non-toxic and abundantly available in Earth's crust. Silicon PV modules have shown long-term stability over decades.

Silicon solar cells are getting cheaper with time, too. Main factors contributing to this decrease in price are efforts to develop new methods and interesting designs of cell architecture, growing automation of factories and use of diamond wire saws to cut silicon wafers, thereby reducing the cutting loss of costly silicon ingot.

Finally, cell manufacturing equipment are becoming increasingly efficient, reducing power consumption. At present, price of modules is about US$ 0.30 per watt DC, which is speculated to fall to US$ 0.18 per watt DC in the next five years—a forty per cent drop.

There is not much difference in price between imported and Indian solar panels. At present in India, just the panel costs about ₹ 45 per watt of peak power generated for monocrystalline cells, ₹ 30 per watt for polycrystalline cells and ₹ 25 per watt (approximately) for amorphous silicon cells.

There are mainly three types of silicon solar cells, as explained below and shown in Fig. 2.

Monocrystalline silicon solar cells. These solar cells are made from a very pure type of silicon. Monocrystalline cells are cut from a silicon ingot grown from a single large crystal of silicon. The higher the purity, the better the alignment of the molecules, making the device more efficient at converting sunlight into electricity.

Monocrystalline solar cells are the most efficient of all. Their efficiencies have been documented at upwards of 20 per cent (25 per cent maximum in a laboratory). These also last the longest, more than 25 years, but these are the most expensive, too. One of the reasons why these are expensive is that the four-sided cutting process ends up wasting a lot of silicon, sometimes more than half.

Polycrystalline silicon solar cells. These solar cells are cut from cast square ingots made by melting many smaller crystals. Liquid silicon is poured into blocks, which are cut into thin plates. This way these are made much more affordable since hardly any silicon is wasted during manufacturing.

Because there are many crystals in each cell, there is less freedom for electrons to move. As a result, these cells have lower efficiency ratings than monocrystalline panels. These cells are large—a factor that must be considered if there is a space constraint. On average, conversion efficiency is between 14 and 16 per cent with laboratory records over 21 per cent.

Another drawback is that these cells have lower heat tolerance. Their life is more than twenty years.

Amorphous silicon solar cells. The word amorphous means shapeless. Amorphous silicon is the non-crystalline form of silicon. Amorphous silicon panels are formed by vapour—depositing a thin layer of silicon material (about one micrometre thick) on a substrate material such as glass, metal or plastic at temperature as low as 75°C. These are either tandem or triple-layer systems that are optimised to capture light from the full solar spectrum. These are cost-competitive. Since amorphous silicon is full of natural defects, its efficiency is low, about seven to eight per cent. Life